

Defense sheet — Exercise set 2

This set has two problems, each in its own file: - **Exercise 1 — Least-Squares Monte Carlo (LSM / Longstaff-Schwartz):** `exercisel_def (2).py` - **Exercise 2 — Finite Difference (FTCS) for European and American puts:** `exercise2_def (2).py`

Line numbers below refer to those two files. Where a “why” requires my own reasoning, it is left as a TODO (Bruno) for me to answer out loud.

The task (in my own words)

Exercise 1 — LSM (Longstaff-Schwartz, 2001). - Part (i): reproduce the textbook 8-path numerical example from the paper. Strike $K = 1.10$, $r = 6\%$, three yearly exercise dates ($dt = 1$, $T = 3$). Print the cash-flow matrix at each time, the in-the-money regression tables, the fitted $E[Y|X] = a_0 + a_1 X + a_2 X^2$, the optimal early-exercise decisions, the stopping rule, the option cash-flow matrix, and finally the American and European put prices to 5 decimals. - Part (ii): replace the 8 hard-coded paths with $N = 100,000$ simulated GBM paths (50k + 50k antithetic), $K = 40$, $M = 50$ exercise dates per year, and replicate the “red box” columns of Table 1 of the paper (Simulated American, s.e., closed-form European, early-exercise value) for the 20 (S_0 , σ , T) combinations. Here the basis is the first three weighted Laguerre polynomials evaluated at $x = S/K$.

Exercise 2 — Finite Difference (FTCS). - Part (i): adapt a provided `ftcs_bs_put.py` to price European puts via an explicit (forward-time, central-space) scheme in log-price space $m = \log S$, for the same 20 parameter combinations, and compare against closed-form Black-Scholes. - Part (ii): turn the European FTCS solver into an American one by adding the early-exercise (free-boundary) projection $U = \max(U, \text{intrinsic})$ after each time step. Compare FD-American against the LSM-American of Exercise 1 as a correctness check. - Part (iii): assemble the full “outside the red box” columns of Table 1 (FD American, closed-form European, FD early-exercise value, and the FD-minus-LSM difference in early-exercise value).

TODO (Bruno): In one sentence, why does the American put price exceed the European put price here, and why is that *not* true for an American call on a non-dividend stock?

My approach and why

Exercise 1 — LSM. - Backward induction over exercise dates. At each date, regress discounted realized future cash flows Y on a basis in the current spot X , but **only over in-the-money paths**, then compare immediate exercise against the fitted continuation value. - Part (i) uses a degree-2 monomial basis $\{1, X, X^2\}$ via an sklearn pipeline (`exercisel_def (2).py:107`). - Part (ii) uses the first three weighted Laguerre polynomials in $x = S/K$ (`exercisel_def (2).py:185-191`). - Variance reduction by antithetic paths (`exercisel_def (2).py:177-178`).

TODO (Bruno): Why regress only on in-the-money paths rather than all paths? (Out-of-the-money continuation values are irrelevant to the exercise decision — but say *why* in your own words and tie it to the paper.)

TODO (Bruno): Why monomials $\{1, X, X^2\}$ in Part (i) but weighted Laguerre in Part (ii)? Was this just “match the paper,” or numerical conditioning? Be ready to defend both.

TODO (Bruno): Antithetic variates: why does pairing Z with $-Z$ reduce variance for this payoff, and what property of the estimator makes the variance reduction work? State the covariance argument.

Exercise 2 — FTCS. - Work in log-price $m = \log S$ so the Black-Scholes PDE has constant coefficients, build the explicit evolution matrix F once per parameter set, and march forward in transformed time (time-to-maturity). - American: project onto the payoff after each step (`exercise2_def (2).py:105`).

TODO (Bruno): Why transform to log-price $m = \log S$ before discretizing instead of working directly in S ? (Constant coefficients / uniform grid — but explain the consequence for the matrices T_1, T_2 .)

TODO (Bruno): FTCS is *explicit*. Why is that an acceptable choice here, and what stability condition are you implicitly relying on with $N_t = 40000 \cdot T$? (See “hardest question” below — this is the danger zone.)

Key code sections (file + what it does + why I wrote it that way)

Exercise 1 — `exercisel_def (2).py`

- **:11-24** — Part (i) parameters and the 8 hard-coded paths from the paper. C (cash-flow matrix) initialized to zeros at :27, shape (8, 3).
- **:32-86** — three printing helpers (`print_formatted_table`, `print_regression_table`, `optimal_table`) that produce the per-time tables shown in the report. Pure presentation, no pricing logic.
- **:89 LSQM(S,C,dt,T,K,r)** — the Part (i) LSM engine.
 - :91 terminal payoff $C[:, T-1] = \max(K - S[:, T], 0)$.
 - :93 `C_european` keeps a copy of the terminal payoff for the European price later.
 - :96 backward loop $i = T-2 \dots 0$.
 - :98 regressor $X = S[:, i+1]$ masked by $<K$ (so OTM entries are zeroed before the ITM filter), reshaped to 2D.
 - :99-101 builds $Y = \text{sum of all future cash flows discounted back to time } i+1 \text{ using discount_factors} = \exp(-r \cdot dt \cdot (u-i))$ with $u = \text{arange}(i+1, \text{ncols})$.
 - :102-104 ITM mask $S[:, i+1] < K$; restrict X, Y to ITM.
 - :107-108 degree-2 polynomial + linear regression pipeline, fit on ITM data.
 - :111-112 extracts intercept a_0 and coefficients a_1, a_2 .
 - :117-120 continuation = model prediction on ITM paths; `exercise = K - S` on ITM paths.
 - :122-127 exercise rule: if `exercise > continuation`, set $C[j, i]$ to exercise value and **zero out all later cash flows** $C[j, i+1:] = 0$; else $C[j, i] = 0$.
 - :131 stopping matrix from $C \neq 0$.
 - :147-152 American price: discount each path's cash flows by $\exp(-r \cdot dt \cdot t)$ for $t=1..T$, sum per path, average.
 - :156 European price: $\text{mean}(C_european * \exp(-r \cdot dt \cdot T))$.
- **:167-173** — Part (ii) parameters ($K=40, M=50, N=100000$).
- **:175-183 simulate_paths** — GBM exact-step simulation; :177 draws $N//2$ normals, :178 stacks $-Z$ for antithetics; :181-182 exact log-Euler update $S[:, i+1] = S[:, i] \cdot \exp((r - 0.5 \sigma^2) dt + \sigma \sqrt{dt} Z)$.
- **:185-191 laguerre_basis** — $x=S/K$; $L_0=\exp(-x/2)$, $L_1=\exp(-x/2)(1-x)$, $L_2=\exp(-x/2)(1-2x+x^2/2)$; stacked column-wise.
- **:193-196 black_scholes_put** — closed-form European put.
- **:198-235 LSM_pricing** — Part (ii) LSM engine; same backward structure as LSQM but basis is `laguerre_basis (:212)` and `LinearRegression()` with default intercept (:215). Returns per-path discounted cash flows.
- **:237-267** — loop over the 20 (S_0, σ, T) combos: $M_total = M \cdot T$, $dt = 1/M$, `simulate`, `price`, `compute american = mean(cashflow)`, `european = black_scholes_put(...)`, `ee = |american - european| (:257)`, `se = std(cashflow)/sqrt(N) (:258)`.

TODO (Bruno): :91 and :203 index $S[:, T] / S[:, M_total]$ for the terminal payoff while the cash-flow matrix C has only T (resp. M_total) columns indexed $0..T-1$. Explain the off-by-one convention: S has one more column (the $t=0$ spot) than C . Be ready to point at exactly which column of S is “today” and which is maturity.

TODO (Bruno): :257 uses `ee = np.abs(american - european)` (absolute value) in the Part (ii) block of `exercisel_def`, but the copy in `exercise2_def (2).py`:223 uses the *signed* `ee = american_value -`

european_value. Which one matches the paper’s “early exercise value” column, and is the absolute value ever wrong (can LSM-American dip below European by noise)?

TODO (Bruno): Standard error $se = \text{std}(\text{cashflow})/\sqrt{N}$ at :258. With antithetic sampling the N draws are **not independent** (each path is paired with its antithetic). Is dividing by $\sqrt{N}$ the correct s.e., or does it understate/overstate it? Defend your formula.

Exercise 2 — exercise2_def (2).py

- :5-14 — parameters $K=40$, $r=0.06$, and the 20 (S_0 , σ , T) combos.
- :18-64 — **Part (i) European FTCS** loop:
 - :24-26 closed-form BS put for reference.
 - :29 domain length $L = 2 \cdot \log(K) + 2$; :30 $N_x = 1000$; :31 $N_t = 40000 \cdot T$; :32-33 steps $h = L/N_x$, $k = T/N_t$.
 - :35-36 central-difference matrices: T_1 (first derivative, $+1/-1$ off-diagonals), T_2 (second derivative, -2 on diagonal, 1 off-diagonals), size (N_x-1) .
 - :38-40 explicit evolution matrix $F = (1 - r k) I + 0.5 k \sigma^2 / h^2 * T_2 + k (r - 0.5 \sigma^2)/(2h) * T_1$.
 - :42-43 grid $mvec$, centered and shifted by $+\log(K)$.
 - :45-46 initial condition $U[:,0] = \max(K - \exp(mvec), 0)$ (payoff as the time-0 condition in time-to-maturity).
 - :48-52 forward march; :51 boundary correction term $p[0]$ injected at the low- m (deep ITM) boundary node, scaled by $K \cdot \exp(-r \cdot \text{time2mat})$.
 - :54 np.interp to read off the price at $\log(S_0)$.
- :66-118 — **Part (ii) American FTCS** loop: identical to Part (i) except $\text{intrinsic} = \max(K - \exp(mvec), 0)$ (:95) and the single added projection line :105 $U[:,i+1] = \text{np.maximum}(U[:,i+1], \text{intrinsic})$. $ee = \text{ftcs_american} - \text{bs_put}(:108)$.
- :120-204 — re-imports and re-runs the **Exercise 1 Part (ii) LSM** code (paths, Laguerre basis, BS put, LSM_pricing) to regenerate results_lsmc for the cross-check.
- :235-248 — correctness test table: $|\text{FD American} - \text{LSM American}|$ per row.
- :250-261 — full “outside the red box” table including the signed $\text{FD EE} - \text{LSM EE}$ difference.

TODO (Bruno): :29 $L = 2 \cdot \log(K) + 2$. Why this width, and why center the grid at $\log(K)$ (:43)? What goes wrong at short maturity / high vol if the domain is too narrow — and what is the analogy with the COS truncation-range $[a,b]$ issue I hit in Assignment 2?

TODO (Bruno): :51 the boundary term $p[0]$. Explain in words what Dirichlet boundary condition this enforces at the deep-in-the-money edge, why it carries the factor $K \cdot \exp(-r \cdot \text{time2mat})$, and what happens at the *other* (high- S) boundary where no correction is added.

TODO (Bruno): :105 is described in the report as “the single line that changes.” Explain why projecting $U = \max(U, \text{intrinsic})$ after each explicit step is a valid (operator-splitting / explicit-penalty) treatment of the American free boundary, and what its accuracy cost is versus a proper LCP/PSOR solve.

Design decisions I must justify

1. **ITM-only regression** (:103-104, :211-213). Factual: the mask is $S[:,i+1] < K$. **TODO (Bruno):** justify why this is the Longstaff-Schwartz prescription and not a shortcut.
2. **Basis choice:** monomials degree 2 in Part (i) (:107) vs weighted Laguerre in Part (ii) (:185-191). Factual: the report says normalization $x=S/K$ avoids underflow in $\exp(-x/2)$ for S in $[36,44]$. **TODO (Bruno):** confirm you can derive the underflow argument and say why monomials were fine for the small $K=1.10$ example.
3. **Antithetic variates** (:177-178). **TODO (Bruno):** justify and quantify expected variance reduction.

4. **Exact GBM step** (:182) rather than Euler. **TODO (Bruno):** why is the log update exact for GBM, and does discretization bias still enter the American price through the discrete exercise dates regardless?
5. **M = 50 exercise dates/year as a proxy for continuous American exercise** (:172, :251). **TODO (Bruno):** why is a Bermudan with 50/100 dates a good stand-in for the true American, and which direction is the bias?
6. **Explicit FTCS with Nt = 40000*T** (exercise2_def (2).py:31). **TODO (Bruno):** show that this choice keeps the scheme inside the explicit stability region (compute the CFL-type bound; see hardest question).
7. **np.interp to extract the price at log(S0)** (:54, :107). Factual: linear interpolation between grid nodes. **TODO (Bruno):** is linear interpolation accurate enough given h, or should it be higher order?

Results and what they mean

Exercise 1, Part (i) (from the report): American put price **0.11443**, European put price **0.05638** on the same 8 paths. The fitted regressions match the paper: at $t=2$, $E[Y|X] = -1.06999 + 2.98341 X - 1.81358 X^2$; at $t=1$, $E[Y|X] = 2.03751 - 3.33544 X + 1.35646 X^2$. Stopping rule and option cash-flow matrix reproduce the paper's table.

Exercise 1, Part (ii) (report Fig. 13): the 20-row replication of the red-box columns; e.g. $S=36$, $\sigma=0.2$, $T=1$ gives Simulated American **4.4714 (s.e. 0.0094)**, closed-form European **3.8443**, early-exercise value **0.6271**. s.e. all in the 0.006–0.023 range.

Exercise 2, Part (i) (report Fig. 14): FD-European vs BS-European agree to $\sim 1e-4$ across all 20 rows (largest $|Difference| = 0.0008$ at $S=40$, $\sigma=0.2$, $T=1$).

Exercise 2, Part (ii) (report Fig. 16): FD-American prices, all above the corresponding European, with early-exercise values matching the LSM ones in sign and magnitude.

Exercise 2 cross-check (report Fig. 17): $|FD\ American - LSM\ American|$ is below 0.05 for all 20 rows (max ~ 0.016), and the Part (iii) FD EE - LSM EE differences are small and alternate in sign.

TODO (Bruno): The FD-European prices are *systematically slightly below* BS (every Difference is negative, $\sim -1e-4$ to $-8e-4$). Is that a bias of the explicit scheme, the truncated domain, or the interpolation? Pick one and defend it — Fang will ask why the sign is consistent.

TODO (Bruno): The FD-American vs LSM-American differences are ~ 0.005 – 0.016 , an order of magnitude larger than the FD-European vs BS error ($\sim 1e-4$). Why is the American comparison so much looser? (Hint: LSM bias + 50-date Bermudan + regression noise vs a near-exact PDE.)

The hardest question Fang could ask here — and my answer

Likely hardest: “Your FTCS scheme is explicit. Is it stable for $N_x = 1000$, $N_t = 40000*T$? Derive the condition and check it for your worst-case $\sigma = 0.4$.”

Factual setup from the code: in log-space the diffusion coefficient is $0.5*\sigma^2$, $h = L/N_x$ with $L = 2 \log K + 2 = 2 \log 40 + 2 \approx 9.376$, so $h \approx 0.00938$; $k = T/N_t = 1/40000 = 2.5e-5$ per unit time.

The explicit-scheme stability constraint for the diffusion part is roughly $0.5*\sigma^2 * k / h^2 \leq 1/2$, i.e. $k \leq h^2 / \sigma^2$.

TODO (Bruno): Plug in the numbers: $h^2 / \sigma^2 = (0.00938)^2 / 0.16 \approx 5.5e-4$, and $k = 2.5e-5 \ll 5.5e-4$, so the diffusion number $\approx 0.045 \ll 0.5$. State this out loud, confirm the scheme is comfortably stable, and explain *why the author picked Nt so large* (was it stability, or matching the paper's accuracy?). Also be ready to say what the $(1 - r k)$ reaction term and the

T1 advection term contribute to the stability bound — and whether the advection (first-derivative) term can cause spurious oscillations even when the diffusion bound is met.

Backup hardest (LSM): “Is your LSM price biased high or low, and why?” TODO (Bruno): Standard result — using the *same* paths to estimate the continuation regression and to value the option introduces a look-ahead/in-sample bias in the continuation estimate, so the exercise decision uses noisy information. State the direction of the resulting price bias and how an out-of-sample / two-pass scheme would fix it.

“Show me in your code where you do X” — anticipated spots

- “...filter to in-the-money paths before regressing.” `exercisel_def (2).py:102-104` (`itm_indexes = S[:,i+1] < K`), and :211-213 for Part (ii).
- “...build the discounted continuation target Y.” :99-101 (Part i) and :207-209 / :182-184 (Part ii).
- “...fit the continuation-value regression and read off the coefficients.” :107-112 (monomial pipeline) and :215-217 (Laguerre).
- “...make the exercise-vs-continue decision and zero out future cash flows.” :122-127 (Part i), :224-229 / :194-199 (Part ii).
- “...compute the American and European prices.” :147-157 (Part i American + European).
- “...generate antithetic paths.” :177-178.
- “...evaluate the weighted Laguerre basis.” :185-191.
- “...the closed-form Black-Scholes put.” :193-196 (Ex.1), `exercise2_def (2).py:24-26` and :74-76.
- “...assemble the explicit FTCS evolution matrix F.” `exercise2_def (2).py:35-40` (and :85-90).
- “...set the initial/boundary conditions of the PDE.” initial: :45-46; boundary term: :51 (and :103).
- “...the one line that makes it American.” `exercise2_def (2).py:105` (`U = np.maximum(U, intrinsic)`).
- “...read the price off the grid at S0.” :54 and :107 (`np.interp(np.log(S0), mvec, U[:, Nt])`).
- “...cross-check FD against LSM.” :235-248 (`|FD American - LSM American|`).